

Lab Assignment 4

Name: Writick Parui

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Course: PCS221 – Cloud Computing

Platform: Microsoft Azure

Q1 – Azure Virtual Machine Web Server Deployment

Objective

To create an Ubuntu Virtual Machine on Microsoft Azure, install and configure Apache Web Server, allow HTTP traffic on port 80, and host a webpage accessible via the VM's Public IP address.

Step 1 – Virtual Machine Creation

The following configuration was used to create the VM on Azure Portal:

VM Name: pcs221-webserver

Operating System: Ubuntu Server 22.04 LTS

VM Size: Standard_B1s

Region: Central India

Authentication: Password

Username: azureuser

Public IP Name: pcs221-web-ip

Virtual Network: pcs221-vnet

VM was deployed successfully and showed running status on the Azure Portal.

Step 2 – SSH Connection to VM

Connected to the VM using the following command:

```
ssh azureuser@20.204.118.73
```

SSH connection was established successfully from the terminal.

Step 3 – Apache Web Server Installation

Ran the following commands on the VM to install Apache:

```
sudo apt update
sudo apt install apache2 -y
```

Apache installed successfully without any errors.

Step 4 – Starting and Verifying Apache Service

Started and enabled Apache to run on boot:

```
sudo systemctl start apache2
sudo systemctl enable apache2
sudo systemctl status apache2
```

Output received:

```
apache2.service - The Apache HTTP Server
```

Loaded: loaded (/lib/systemd/system/apache2.service)
Active: active (running)

Step 5 – Configure HTTP Networking (Port 80)

Added an inbound rule in the Network Security Group (NSG) to allow HTTP traffic:

Port 22 – TCP – SSH Access

Port 80 – TCP – HTTP Web Access

HTTP access was enabled successfully.

Step 6 – Creating and Hosting index.html

Created the webpage at /var/www/html/index.html with the following content:

```
<!DOCTYPE html>
<html>
<head>
  <title>PCS221 Cloud Computing</title>
  <style>
    body { font-family: Arial; background-color: #f2f2f2; padding: 40px; }
    .container { background: white; padding: 30px; border-radius: 10px; }
  </style>
</head>
<body>
  <div class="container">
    <h1>Writick Parui</h1>
    <p><b>Roll Number:</b> 8025320111</p>
    <p><b>Course:</b> PCS221 Cloud Computing</p>
    <h2>What I Learned About Cloud Computing</h2>
    <p>Cloud Computing allows users to access scalable computing
    resources through the internet. It simplifies infrastructure
    management, improves scalability, and reduces hardware dependency.</p>
    <a href="contact.html">Go to Contact Page</a>
  </div>
</body>
</html>
```

The webpage was accessed using the public IP in the browser:

http://20.204.118.73

Website opened and loaded successfully in the browser.

Q2 – Contact Page Deployment

Objective

Create a second webpage named contact.html on the same VM containing an email address, a profile photo, and two-way navigation links between index.html and contact.html.

contact.html Code

```
<!DOCTYPE html>
<html>
<head>
  <title>Contact Page</title>
  <style>
    body { font-family: Arial; background: #eef2ff; padding: 40px; }
    .card { background: white; padding: 30px; border-radius: 10px; }
    img { border-radius: 10px; }
  </style>
</head>
<body>
  <div class="card">
    <h1>Contact Information</h1>
    <p><b>Name:</b> Writick Parui</p>
    <p><b>Roll No:</b> 8025320111</p>
    <p><b>Email:</b> writick2210@gmail.com</p>
    
    <br><br>
    <a href="index.html">Back to Home Page</a>
  </div>
</body>
</html>
```

Navigation Verification

index.html to contact.html: Working

contact.html to index.html: Working

Both pages are accessible from each other via the navigation links. The contact page displays the email address and profile photo correctly.

Q3 – Static Website Hosting via Azure Storage Account

Objective

Host a static website using Microsoft Azure Storage Account without using a Virtual Machine.

Step 1 – Storage Account Creation

Created a new Storage Account with the following configuration:

Storage Account Name: pcs221storageweb
Performance: Standard
Replication: LRS (Locally Redundant Storage)
Account Kind: StorageV2
Region: Central India

Storage Account created successfully.

Step 2 – Enable Static Website Feature

Navigated to Data Management > Static Website and enabled it with the following settings:

Static Website: Enabled
Index Document: index.html
Error Document: error.html
Primary Endpoint: https://pcs221storageweb.z29.web.core.windows.net

Azure automatically created the \$web container after enabling this feature.

Step 3 – index.html for Static Website

```
<!DOCTYPE html>
<html>
<head>
  <title>Azure Static Website</title>
</head>
<body>
  <h1>Writick Parui</h1>
  <p>Roll Number: 8025320111</p>
  <p>Course: PCS221 Cloud Computing</p>
  <p>This website is hosted using Azure Storage without a VM.</p>
</body>
</html>
```

error.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Error Page</title>
```

```
</head>
<body>
  <h1>404 Error</h1>
  <p>Requested page not found.</p>
</body>
</html>
```

Step 4 – Upload Files to \$web Container

Both index.html and error.html were uploaded to the \$web container via the Azure Portal.

Files uploaded successfully to the \$web container.

Step 5 – Website Verification

Accessed the website using the generated static website URL:

`https://pcs221storageweb.z29.web.core.windows.net`

Website loaded successfully and displayed the name, roll number, course, and required message.

GitHub Repository

The following files were pushed to the GitHub repository:

- index.html
- contact.html
- error.html
- public_ip.txt – contains: 20.204.118.73

Conclusion

This assignment gave practical experience with deploying web applications on Microsoft Azure. In Q1, an Ubuntu VM was set up with Apache and a live webpage was hosted using the public IP. In Q2, a contact page was added with navigation between both pages. In Q3, a static website was hosted using Azure Blob Storage without needing a VM at all. Overall the assignment covered VM setup, web server configuration, NSG rules, and Azure Storage-based hosting.

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